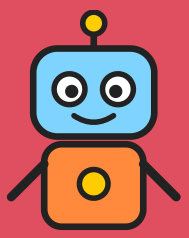
Robotics Project #11 · Kids.JoinDev

Obstacle-Spotting Robot

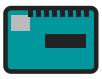
Ages 10+ · 90 min · Build with a grown-up!



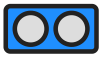
🎯 Goal

Give a robot bat-like 'ears' that measure distance and turn its head away from walls.

🧰 You need



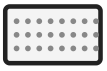
Arduino Uno



HC-SR04 distance sensor



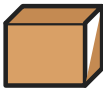
SG90 micro servo



breadboard



jumper wires



cardboard robot head

🔧 Build it

1. Wire the sensor: VCC to 5V, GND to GND, TRIG to pin 9, ECHO to pin 10. ☐
2. Wire the servo: brown GND, red 5V, orange to pin 6. ☐
3. Tape the sensor on the servo arm: this is the head. ☐
4. Upload obstacle.ino. ☐
5. Move your hand close: the head looks away! ☐
6. Open Serial Monitor to see the distance in cm. ☐

⚠️ Safety

Unplug USB before rewiring. Keep fingers clear of the servo while it moves.

💡 How it works

The sensor sends a squeak too high to hear and times the echo, like a bat. Short time = close object.

Code: obstacle.ino (in the project folder)